Steganography Using C++ (LSB Technique)

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Abstract
This project implements image steganography using the Least Significant Bit (LSB) technique in C++. It hides secret text inside a BMP image and retrieves it without visibly altering the image.

Objective
To securely embed and extract secret messages from BMP images using LSB substitution.

Tools and Technologies
Language: C++
Compiler: g++
Platform: Linux/Ubuntu
Editor: Visual Studio Code

Modules
• Encoding
• Decoding
• BMP Validation
• Secret File Handling
• Command-line Processing

Algorithm
Encoding: Read BMP image and secret file, replace image byte LSBs with secret bits, and create a stego image.

Decoding: Read the stego image, extract LSBs, reconstruct the hidden message, and display it.

Features

• Hides secret text inside BMP images
• Recovers hidden messages accurately
• Preserves image quality
• Easy command-line usage

Applications

Secure communication, Digital watermarking, Copyright protection, Confidential information sharing.

Conclusion

The project successfully demonstrates image steganography using the LSB technique and provides a simple, reliable method for hiding and retrieving text from BMP images.